

Mousetrap cascades: mean-field theory vs simulation

When does a chain reaction of ping-pong balls take off, in one and two dimensions?

The question

Setup

Mousetraps with balls sit on a lattice at density ρ . A fired trap launches balls that may spring other traps.

Question

Below which density does the reaction fizzle out, and above it, what fraction of traps ends up fired?

Approach

A mean-field theory with one parameter, κ , tested against lattice simulations in 2D and 1D.

What each notebook does

Notebook	Contents
Notebook 0	Builds the 2D lattice model step by step
Notebook 1	1D model; why the default parameters die out; what sustains a cascade
Notebook 2	Background theory: R_0 , critical density, reaction-diffusion
Notebook 3	2D: derives κ , onset and final size, then tests them against simulation
Notebook 4	1D: the same three tests on a line
Both 3 and 4	Last sections (3.9, 4.9): the supercritical test, with $m = 5$

Run each notebook top to bottom, with the function files in the same folder.

Mean-field theory

Branching number

$$\kappa$$

Mean successful activation attempts per fired trap.

Onset density

$$\rho = 1/\kappa$$

The cascade can only grow when $\rho\kappa > 1$.

Final size

$$F = 1 - e^{-\rho\kappa F}$$

Nontrivial root $F > 0$ only if $\rho\kappa > 1$.

Below onset $F = 0$: a few traps fire, then the chain dies. Above it, a finite fraction of the system fires.

Where the final size comes from

Equation

$$F = 1 - \exp(-\rho\kappa F)$$

Each site receives a Poisson number of attempts with mean $\kappa\rho F$.

Threshold

The slope of the right-hand side at $F = 0$ is $\rho\kappa$, so a non-zero solution exists only when $\rho\kappa > 1$. The onset density is $1/\kappa$.

Computing κ from the model

Form	κ	p onset	Matches simulator
Additive (PDF eq. 3)	0.9942	1.0058	No
Mixture (used)	0.4971	2.0116	Yes

Mixture: $\kappa = m[(1 - p_{\text{long}}) \cdot q_s + p_{\text{long}} \cdot E[q_\ell(k)]]$ with short-range 0.4500 and long-range 0.0471 contributions at the default parameters ($m = 1$).

Defaults: $p_{\text{succ_short}}$ 0.9, p_{long} 0.5, $p_{\text{succ_long}}(k) = 0.2 \cdot \exp(-k/5)$, $f_k = \text{geometric}(0.3) + 1$

Three comparisons

1

First generation

Newly fired traps after one step against $\rho\kappa$, with z-scores and a fitted slope.

2

Final fired fraction

Mean simulated fraction against $F(\rho)$, with standard errors and RMSE across lattice sizes L .

3

$P(\text{macroscopic})$

Where the probability of a large cascade crosses 50%, for several thresholds and L .

Check 1 validates κ itself. Checks 2 and 3 test the macroscopic predictions.

At default parameters κ is validated in 2D and 1D

2D

0.492

fitted κ against 0.497 predicted, $|z| \leq 1.59$

1D

0.493

fitted κ against 0.497 predicted, $|z| \leq 0.45$

First-generation counts follow $\rho\kappa$ within sampling error in both geometries.

Default parameters: subcritical, and theory agrees

Size estimate

$$1 / ((1 - \rho\kappa) \cdot \rho N)$$

Fired fraction from one seed,
matching the simulated tables.

2D agreement

RMSE 0.0017

at $L = 50$, falling to 0.0004 at $L = 100$.

Macroscopic cascades

$\leq 1\%$

1D probability at the loosest
threshold, 0% for the others.

Supercritical test: $\kappa = 4.5$

Parameters

$m = 5$ attempts per fired trap,
all long-range ($p_{\text{long}} = 1$),
success probability 0.9 at
every distance.

Theory

$\kappa = 5 \times 0.9 = 4.5$, onset at $\rho = 0.222$, and a final fired fraction of 98.8% at $\rho = 1$.

Why long-range

Short-range attempts only reach the few cells next to the trap, and most of those are already spent.

With $m = 1$ no density is supercritical, so the earlier checks could not test the onset or $F(\rho)$.

κ counts attempts, not new traps

2D fitted κ

4.43

1.4% below 4.5

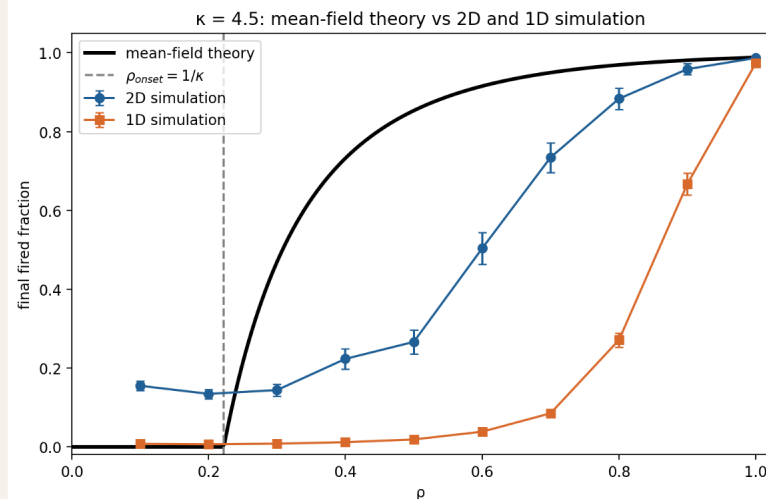
1D fitted κ

3.87

14% below 4.5

The simulator merges several hits on one trap in a generation. In 1D a hop at distance k has only 2 target sites, so the five attempts often coincide. The expected number of distinct targets is 3.85, matching the 1D fit.

Both simulations fall below the theory



2D: 0.27 against 0.85 predicted at $\rho = 0.5$, 0.88 at $\rho = 0.8$, and 0.99 at $\rho = 1$ where theory and simulation meet.

1D: 0.02 against 0.85 at $\rho = 0.5$, 0.27 at $\rho = 0.8$, and 0.97 at $\rho = 1$.

The gap is much larger in 1D.

The onset sits above $1/\kappa = 0.22$

Lattice	Onset, 5% threshold	Onset, 10% threshold	Offset from $1/\kappa$
2D, L = 50	0.32	0.32	+0.10
2D, L = 100	0.32	0.33	+0.10
1D, L = 1000	0.57	0.66	+0.35 to +0.44
1D, L = 2000	0.64	0.73	+0.42 to +0.51

The 2D offset is stable across threshold and size. The 1D onset moves right with both, so it is not a clean threshold there.

Likely cause: attempts are local

Theory assumes

Every attempt lands on a uniformly random site of the whole lattice.

Simulation does

Lands within about 4 sites (mean $k = 4.3$), so the cascade is a front and cells behind it are already spent.

Why 1D is worse

The site a trap was fired from is always one of its two nearest neighbours. In 2D far more fresh sites are in reach.

Expectation and findings

Default parameters: $\kappa = 0.497$

Expected: onset at $\rho = 2.01$, so no density gives a macroscopic cascade.

Found: fitted κ 0.492 (2D) and 0.493 (1D). Final fraction RMSE 0.0017 at $L = 50$ and 0.0004 at $L = 100$ in 2D. Theory holds.

Supercritical: $\kappa = 4.5$

Expected: onset at $\rho = 0.222$ and 98.8% of traps fired at $\rho = 1$.

Found, 2D: below theory until $\rho \approx 0.9$ (0.27 against 0.85 at $\rho = 0.5$), and meets it at $\rho = 1$.

Supercritical, 1D

Found: fired fraction 0.02 against 0.85 predicted at $\rho = 0.5$, and 0.27 at $\rho = 0.8$. The gap is much larger than in 2D.

Where to take the experiment

Test the explanation

- Draw the hop distance k uniformly over the whole lattice and check whether the mean-field curve returns.
- Put the measured effective κ (3.85 in 1D) into the final-size equation.

Widen the experiment

- Other values of m , p_{long} and success kernels, not just one supercritical set.
- Larger 1D lattices and more runs, to see whether a transition exists.

Extend the theory

- A model of the cascade front, to get the spatiotemporal behaviour the problem asks for.
- Measure $p_{\text{succ}}(k)$ and the hop distances on real traps to replace the assumed values.

The left column is the most direct test of the current results.

NEXT STEP

Test whether locality explains the gap

Make the hop distance uniform over the whole lattice and check that the mean-field curve returns. Then repeat with other kernels and values of m , and use the effective κ for $m > 1$.